

Voice Anonymization Made Simple: Training-Free Anonymization with Projected Classifier-Free Guidance

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Abstract—Voice anonymization aims to conceal speaker identity while preserving linguistic and paralinguistic information, yet many existing methods require dedicated training. We propose a training-free, two-stage anonymization method for pre-trained flow-matching voice conversion systems. In the condition-construction stage, the source content representation is regenerated under a randomly selected speaker context to reduce residual source-speaker information while preserving linguistic content. In the speech-generation stage, the source speech is used to identify the generation direction associated with the original speaker, and the source-aligned component is suppressed while useful content guidance is retained. The proposed method modifies only inference-time conditioning and guidance, without updating any pretrained parameters. Experiments on VoicePrivacy 2026 Track 1 show that the two-stage method improves speaker privacy while maintaining competitive word error rate and emotion recognition performance¹.

Index Terms—voice anonymization, speaker privacy, voice conversion, flow matching, classifier-free guidance

I. INTRODUCTION

Speech conveys not only linguistic content but also personal information, including identity, health status, emotional state, geographical background, and social characteristics [1]. As speech is increasingly used for authentication, analytics, human–computer interaction, and generative applications, information beyond the intended purpose of collection may be inferred from recorded utterances. Voice privacy therefore concerns limiting unauthorized access to, or inference of, personal information conveyed through speech [1].

This work focuses on voice anonymization, which aims to prevent reliable re-identification of the source speaker while preserving information required by downstream applications, such as linguistic content and prosodic attributes [2]. The task was formalized by the VoicePrivacy Challenge, which provides common datasets, baselines, and evaluation protocols [3], [4]. Subsequent editions introduced increasingly stronger attacker models, including semi-informed attackers [5], [6], emotion-preservation evaluation [7], [8], and, most recently, stronger ASV attackers pretrained on voice-converted speech together with cross-gender and multilingual evaluation [9], [10].

Voice anonymization methods broadly follow signal-processing or neural synthesis approaches. Signal-processing methods modify acoustic properties such as formants or speaking rate, but may distort speech and remain vulnerable to strong attackers [4], [6], [11]–[15].

Neural synthesis methods reconstruct speech from transformed or disentangled representations. ASR+TTS removes speaker identity through text resynthesis but also discards paralinguistic information [16], while more advanced systems preserve prosodic attributes or disentangle content, prosody, and speaker information before anonymization [2], [17]–[24]. Other approaches anonymize general speech representations or discrete speech tokens without explicit speaker embeddings [25]–[28].

Diffusion and flow-matching models have recently emerged as powerful generative frameworks for voice anonymization. DiffVC+ adapts diffusion-based voice conversion for anonymization [29]; F3-VA generates anonymized speaker embeddings using flow matching before conditional speech reconstruction [30]; and DiffAnon reconstructs codec representations from semantic, prosodic, and speaker conditions with classifier-free guidance to control prosody preservation [31]. These methods demonstrate that diffusion and flow matching enable high-quality and controllable speech anonymization.

However, most existing approaches require anonymization-specific training, additional learned components, or retraining when applied to a different voice conversion model. This raises a broader question: can a pretrained flow-matching-based voice conversion model be converted into a voice anonymizer only by changing its inference procedure?

We address this question with a training-free, two-stage anonymization framework for pretrained generative voice conversion systems. The main idea is to intervene at both the representation and generation levels. During condition construction, the source content representation is regenerated under an external speaker context to reduce its dependence on the original speaker. During speech generation, the source speech is used to identify the generation direction associated with the source identity, and the source-aligned component is suppressed while useful content guidance is retained. Both operations are performed only at inference time and require no parameter updates.

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¹Full code and audio samples can be found at https://github.com/xs125-stack/voice_anon_simple

We implement this framework on Seed-VC v2, a state-of-the-art zero-shot voice conversion system [32]. Seed-VC v2 follows a two-stage generation process: it first constructs the conditions required for conversion and then synthesizes speech through conditional flow matching. This structure makes it suitable for applying our two interventions separately. In the first stage, a randomly selected target utterance guides the reconstruction of the source-content condition. In the second stage, source-aware projected classifier-free guidance steers generation away from the original speaker.

Experiments under the VoicePrivacy 2026 Track 1 (English) protocol show that the proposed training-free two-stage intervention achieves a favorable privacy–utility trade-off, improving speaker privacy while maintaining competitive linguistic and emotion utility compared with the baselines.

II. PRELIMINARIES

A. Voice Conversion in Seed-VC v2

Figure 1 shows the released Seed-VC v2 pipeline. Seed-VC is a zero-shot voice conversion system that transfers a source utterance to an unseen reference speaker while preserving linguistic content [32]. The published Seed-VC model uses continuous semantic features, whereas v2 inserts an autoregressive (AR) speech-token model before the conditional flow-matching acoustic model.² Both versions share the same basic idea: the reference speaker is represented by both a global timbre vector and a complete reference segment that provides in-context information [32]. In Figure 1, black arrows denote shared computation and red arrows denote standard VC.

Stage 1: Condition construction. This stage constructs the complete token condition μ , the acoustic prompt $\mathbf{p}$, and the global style embedding $\mathbf{s}$. The complete token condition is not simply a source content representation. It contains a prompt-side part aligned with the reference speech and an AR-generated part for the content to be synthesized.

The source and target utterances are processed by pretrained narrow- and wide-band tokenizers.³ The source narrow-band tokens provide the linguistic sequence, while the target narrow- and wide-band tokens serve as AR context and prefix. The AR model therefore generates a target-conditioned wide-band sequence for the source portion,

$$\hat{\mathbf{z}}_w = G_{\text{AR}}(\mathbf{z}_n^{\text{src}}; \mathbf{z}_n^{\text{tar}}, \mathbf{z}_w^{\text{tar}}), \quad (1)$$

which is converted to the Mel-frame rate as $\mu_{\text{AR}} = H(\hat{\mathbf{z}}_w)$. Here, μ_{AR} describes the source content to be generated, but in a token form already influenced by the target context.

The target wide-band tokens are regulated separately to form the token-prompt condition $\mu_{\text{prm}} = H(\mathbf{z}_w^{\text{tar}})$. The two parts are then concatenated along time as $\mu = \text{Concat}_t(\mu_{\text{prm}}, \mu_{\text{AR}})$. They are regulated separately because the prompt region and generated region must be aligned to different Mel lengths before concatenation.

²https://github.com/Plachtaa/seed-vc/blob/main/inference_v2.py

³<https://github.com/Plachtaa/ASTRAL-quantization>

The same target utterance also provides the acoustic prompt and global style embedding, $\mathbf{p} = M(x^{\text{tar}})$ and $\mathbf{s} = E_s(x^{\text{tar}})$, where $M(\cdot)$ is Mel-spectrogram extraction and $E_s(\cdot)$ is a pre-trained CAM++ encoder [33]. Their roles are complementary: μ_{prm} tells the model what is spoken in the reference segment, $\mathbf{p}$ shows how that segment is realized acoustically, $\mathbf{s}$ gives the overall speaker and timbre, and μ_{AR} specifies the content to generate next.

Together, these conditions form an aligned in-context reference and consistently direct standard VC toward the target speaker [32].

Stage 2: Speech generation. The conditional flow-matching model uses $(\mu, \mathbf{p}, \mathbf{s})$ to generate the converted Mel-spectrogram. Its velocity network is a diffusion transformer (DiT) [34]. At flow time t and state $\mathbf{x}_t$, the estimator $\mathbf{v}_\theta(\mathbf{x}_t, t; \mathbf{p}, \mathbf{s}, \mu)$ defines the ODE $d\mathbf{x}_t/dt = \mathbf{v}_\theta$, which is integrated from Gaussian noise by a fixed-step Euler solver.

For multi-condition classifier-free guidance (CFG), the estimator is evaluated under three condition sets,

$$\begin{aligned} \mathbf{v}_f &= \mathbf{v}_\theta(\mathbf{x}_t, t; \mathbf{p}, \mathbf{s}, \mu), \\ \mathbf{v}_c &= \mathbf{v}_\theta(\mathbf{x}_t, t; \mathbf{0}, \mathbf{0}, \mu), \\ \mathbf{v}_u &= \mathbf{v}_\theta(\mathbf{x}_t, t; \mathbf{0}, \mathbf{0}, \mathbf{0}), \end{aligned} \quad (2)$$

giving the fully conditioned, token-conditioned, and unconditional predictions. Standard VC combines them as

$$\mathbf{v}_{\text{VC}} = \mathbf{v}_f + \gamma_{\text{joint}}(\mathbf{v}_f - \mathbf{v}_u) + \gamma_s(\mathbf{v}_f - \mathbf{v}_c), \quad (3)$$

where γ_{joint} scales the joint guidance contributed by all conditions, including both token/content and reference-speaker information. In contrast, γ_s scales the additional guidance introduced by the acoustic prompt and style embedding. The Euler solver produces a converted Mel-spectrogram, which a pretrained BigVGAN v2 vocoder [35] converts into waveform audio.

B. Built-in Anonymization in Seed-VC v2

Seed-VC v2 also provides a training-free built-in anonymization mode, shown by the green path in Figure 1. It keeps the same two-stage backbone but removes explicit target-timbre conditioning.

Stage 1: Condition construction. The AR model receives the source narrow-band tokens but no target context or prefix,

$$\hat{\mathbf{z}}_w^{\text{builtin}} = G_{\text{AR}}(\mathbf{z}_n^{\text{src}}; \emptyset, \emptyset), \quad \mu_{\text{AR}}^{\text{builtin}} = H(\hat{\mathbf{z}}_w^{\text{builtin}}), \quad (4)$$

where $\emptyset$ denotes an absent AR context or prefix.

The released wrapper still constructs a token-prompt prefix from `target_audio_path`. However, because the acoustic prompt and style embedding are set to zero, this prefix no longer forms an aligned in-context reference pair. Our preliminary tests found little difference between using a source-derived or target-derived prefix. We therefore use the source utterance by default, giving $\mu_{\text{prm}}^{\text{builtin}} = H(\mathbf{z}_w^{\text{src}})$. The complete built-in token condition is $\mu_{\text{builtin}} = \text{Concat}_t(\mu_{\text{prm}}^{\text{builtin}}, \mu_{\text{AR}}^{\text{builtin}})$, with $\mathbf{p} = \mathbf{0}$ and $\mathbf{s} = \mathbf{0}$.

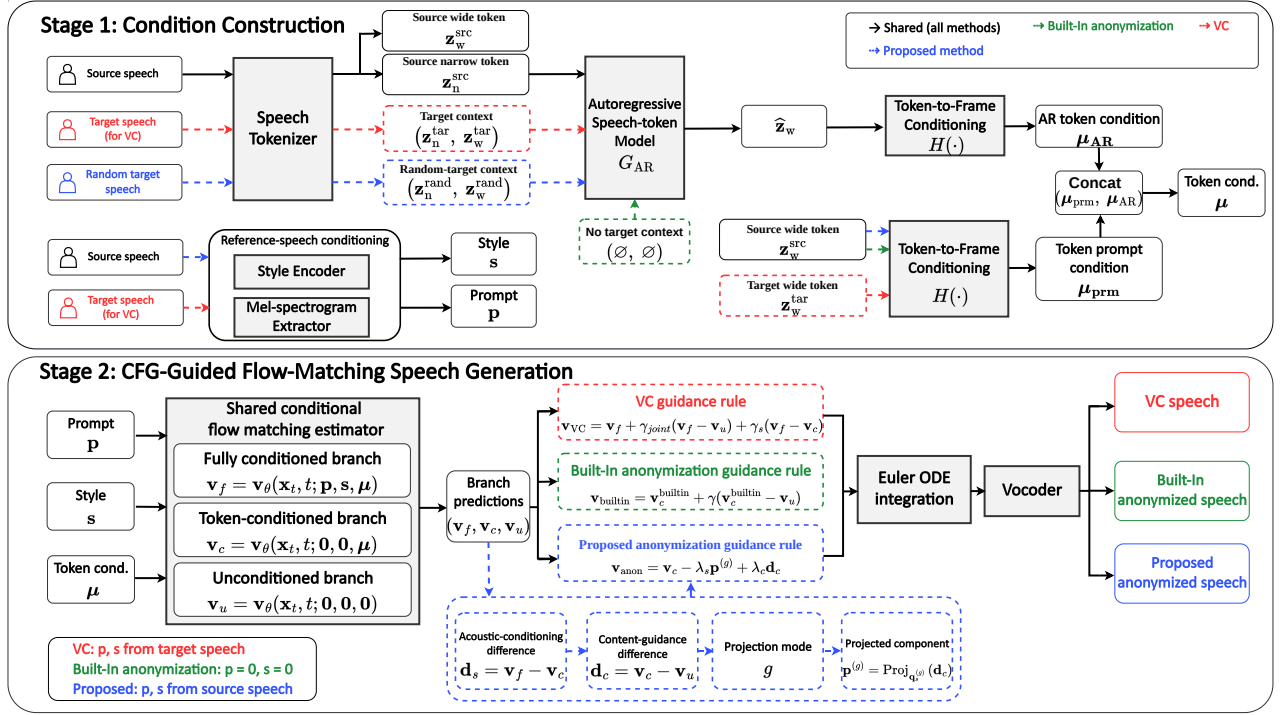


Fig. 1. Overview of standard voice conversion, built-in anonymization, and the proposed training-free method within the shared Seed-VC v2 backbone. The proposed method modifies condition construction to reduce source-speaker dependence and modifies generation guidance to suppress source-aligned information. Detailed condition and guidance definitions are introduced in Sections II and III.

Stage 2: Speech generation. Generation therefore uses only the complete token condition. The estimator is evaluated under the token-conditioned and unconditional settings,

$$\mathbf{v}_c^{\text{builtin}} = \mathbf{v}_\theta(\mathbf{x}_t, t; \mathbf{0}, \mathbf{0}, \mu_{\text{builtin}}), \quad \mathbf{v}_u = \mathbf{v}_\theta(\mathbf{x}_t, t; \mathbf{0}, \mathbf{0}, \mathbf{0}), \quad (5)$$

and combined as

$$\mathbf{v}_{\text{builtin}} = \mathbf{v}_c^{\text{builtin}} + \gamma(\mathbf{v}_c^{\text{builtin}} - \mathbf{v}_u), \quad (6)$$

where γ is the intelligibility CFG scale. The Euler solver integrates this field, and the vocoder renders the anonymized waveform.

III. PROPOSED METHOD

The built-in Seed-VC anonymization mode already provides a training-free way to anonymize speech without updating the pretrained model. However, it still leaves room for improvement in both stages. In Stage 1, the target-related AR inputs are simply removed, so the model does not exploit its ability to regenerate the source content under an alternative speaker context. As a result, the token condition is still largely determined by the source input and may carry residual source-speaker information entangled with the content. In Stage 2, the guidance rule $\mathbf{v}_c^{\text{builtin}} + \gamma(\mathbf{v}_c^{\text{builtin}} - \mathbf{v}_u)$ reduces the direct use of source acoustic conditions, but it neither identifies and removes the source-related information carried in the token guidance nor steers the generation trajectory away from the source speaker. These observations motivate our two modifications: random-target conditioning in Stage 1,

which regenerates the source content under an independently sampled speaker context to weaken residual source-speaker information, and source-conditioning-aware projected CFG in Stage 2, which actively suppresses source-aligned directions during generation. The two modifications are highlighted as the blue path in Figure 1.

A. Stage 1: Random-Target-Conditioned Token and Source-Acoustic Condition Construction

In Stage 1, we change only the AR-generated part of the token condition. As shown in the upper panel of Figure 1, the complete token condition contains two parts: the token-prompt condition μ_{prm} and the AR token condition μ_{AR} . The former describes the prompt region, whereas the latter describes the source content to be generated.

To reduce source-speaker information in the generated source-content tokens, we sample an independent random-target utterance and use its narrow- and wide-band tokens as the AR context and prefix. The source narrow-band tokens still provide the linguistic sequence to be converted. The shared AR model therefore predicts the source-portion wide-band tokens as

$$\hat{\mathbf{z}}_w = G_{\text{AR}}(\mathbf{z}_n^{\text{src}}, \mathbf{z}_n^{\text{rand}}, \mathbf{z}_w^{\text{rand}}). \quad (7)$$

The generated sequence is then converted to the Mel-frame rate as $\mu_{\text{AR}} = H(\hat{\mathbf{z}}_w)$. Thus, μ_{AR} still carries the source linguistic content, but its token realization is guided by an unrelated speaker context.

The token-prompt condition is constructed separately from the source wide-band tokens as $\mu_{\text{prm}} = H(\mathbf{z}_w^{\text{src}})$. We deliberately keep μ_{prm} source-derived because the same source utterance also provides the acoustic prompt $\mathbf{p}$. This keeps the token description of the prompt region aligned with its Mel-spectrogram. If the token-prompt condition were instead taken from the random target while the acoustic prompt remained source-derived, the model would receive random-target prompt tokens together with a source acoustic example. This prompt mismatch differs from the aligned conditioning used during Seed-VC training. In our preliminary tests, such a mismatch sometimes improved speaker separation, but it also made generation unstable and substantially degraded audio quality and increased WER. We therefore use the source utterance for both the token-prompt condition and the acoustic prompt.

The complete token condition is formed by temporal concatenation as $\mu = \text{Concat}_t(\mu_{\text{prm}}, \mu_{\text{AR}})$. This operation is shown explicitly by the *Concat* block in Figure 1. The two parts are first regulated to their own Mel lengths and are concatenated only afterward.

In parallel, the source utterance is passed through the same Mel-spectrogram extractor and CAM++ style encoder used by Seed-VC, giving $\mathbf{p} = M(x^{\text{src}})$ and $\mathbf{s} = E_s(x^{\text{src}})$, where $M(\cdot)$ denotes prompt-Mel extraction and $E_s(\cdot)$ denotes the style encoder. The source-derived μ_{prm} and $\mathbf{p}$ form an aligned prompt example, while $\mathbf{s}$ describes the overall source timbre. However, these source conditions are not used as desired output attributes. They are retained only to expose the effect of source-acoustic conditioning in Stage 2.

Because the complete token condition retains a source prompt part and AR generation starts from source tokens, Stage 1 reduces but not guarantee the removal of all source-speaker information.

B. Stage 2: Source-Conditioning-Aware Projected Classifier-Free Guidance for Conditional Flow-Matching Speech Generation

In Stage 2, we use the source prompt and style embedding only to identify the generation direction associated with the source speaker. At each ODE step, the difference between the fully conditioned and token-conditioned predictions shows how the source acoustic conditions change the current generation state. The difference between the token-conditioned and unconditional predictions shows the effect of the complete token condition μ , which contains both μ_{prm} and μ_{AR} . We project this token-guidance direction onto the source-conditioning direction and suppress the aligned component.

Accordingly, the source prompt $\mathbf{p}$ and style embedding $\mathbf{s}$ serve only as a reference. They identify what should be suppressed rather than what should be reconstructed. The remaining token guidance continues to support linguistic generation, so sampling starts from the token-conditioned field $\mathbf{v}_c$, not from the fully conditioned field $\mathbf{v}_f$.

Concretely, at every ODE step, the shared estimator is evaluated under the same three condition sets as in Eq. (2),

using the $(\mu, \mathbf{p}, \mathbf{s})$ constructed in Stage 1 to obtain $\mathbf{v}_f$, $\mathbf{v}_c$, and $\mathbf{v}_u$. We then form

$$\mathbf{d}_s = \mathbf{v}_f - \mathbf{v}_c, \quad \mathbf{d}_c = \mathbf{v}_c - \mathbf{v}_u, \quad (8)$$

where $\mathbf{d}_s$ is the source-conditioning direction contributed by the source acoustic prompt and style embedding, and $\mathbf{d}_c$ is the guidance direction contributed by the complete token condition. Their alignment identifies the part of token guidance that changes the acoustic state in a way consistent with reintroducing source conditioning. We do not assume that $\mathbf{d}_s$ is a pure speaker direction or that $\mathbf{d}_c$ is a pure content direction.

Both $\mathbf{d}_s$ and $\mathbf{d}_c$ have shape $C \times T$, where C is the feature dimension and T is the number of frames. The projection reference $\mathbf{q}_{s,t}^{(g)}$ is built from $\mathbf{d}_s$ at one of three temporal granularities,

$$\mathbf{q}_{s,t}^{(g)} = \begin{cases} \mathbf{d}_{s,t}, & g = \text{token}, \\ \frac{1}{T} \sum_{\tau=1}^T \mathbf{d}_{s,\tau}, & g = \text{global}, \\ \frac{1}{K} \sum_{\tau \in \mathcal{W}_t} \mathbf{d}_{s,\tau}, & g = \text{smooth}, \end{cases} \quad (9)$$

where *Token* uses the source-conditioning vector at the same frame, *Global* uses one utterance-level mean, and *Smooth* uses a local sliding-window mean over $\mathcal{W}_t = \{t-10, \dots, t+10\}$.⁴ These choices respectively preserve frame-level variation, remove local variation, or provide an intermediate smoothed reference.

The component of token guidance aligned with the selected reference is

$$\mathbf{p}_t^{(g)} = \frac{\langle \mathbf{d}_{c,t}, \mathbf{q}_{s,t}^{(g)} \rangle}{\max\left(\|\mathbf{q}_{s,t}^{(g)}\|_2^2, \epsilon\right)} \mathbf{q}_{s,t}^{(g)}. \quad (10)$$

The vector being projected is always $\mathbf{d}_{c,t}$; only the temporal construction of the reference changes across modes. We treat $\mathbf{p}^{(g)}$ as a proxy for source-speaker information remaining in the token pathway.

The proposed guidance rule removes this component and then reinforces the token guidance,

$$\mathbf{v}_{\text{anon}} = \mathbf{v}_c - \lambda_s \mathbf{p}^{(g)} + \lambda_c \mathbf{d}_c, \quad (11)$$

where λ_s controls source-aligned suppression and λ_c controls token guidance. Using $\mathbf{v}_c$ as the base field prevents the source prompt and style from directly reconstructing the source voice, while the difference between $\mathbf{v}_f$ and $\mathbf{v}_c$ still indicates which direction should be removed. The shared Euler solver then integrates $\mathbf{x}_{t_{k+1}} = \mathbf{x}_{t_k} + \Delta t_k \mathbf{v}_{\text{anon}}(\mathbf{x}_{t_k}, t_k)$, and the pretrained vocoder renders the final waveform.

⁴ $K = 21$: the current frame plus the ten preceding and ten following frames, with reflection padding at utterance boundaries. We therefore refer to this smooth variant as *Smooth-21* in the experiments.

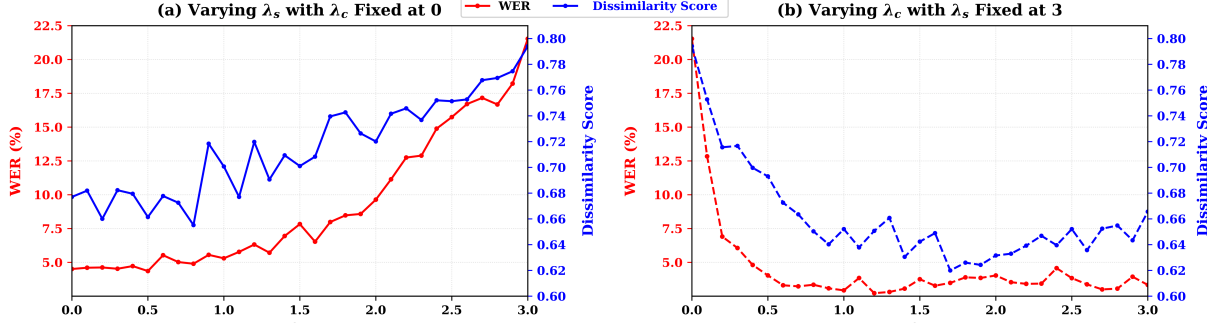


Fig. 2. Effects of λ_s and λ_c on the 30-utterance screening subset. A higher dissimilarity score indicates lower source-speaker similarity, and WER is computed at the corpus level.

IV. EXPERIMENTS

Our experiments proceed in three steps. First, we construct a small screening subset containing 30 utterances selected from the VPC LibriSpeech development clean set [36], denoted as *Libri_dev*.⁵ On this subset, we test whether the two coefficients in Eq. (11) behave as intended: increasing λ_s should strengthen anonymization by removing more source-aligned guidance, whereas increasing λ_c should strengthen content guidance and reduce WER. Second, after confirming these roles, we use the same subset to search for the best (λ_s, λ_c) pair for each combination of random-target condition construction and projection mode. Third, with the selected scale pairs fixed, we evaluate all systems on the full VPC 2026 Track 1 development and evaluation sets. These full-set experiments serve as ablation studies for the random-target-conditioned construction of μ_{AR} introduced in Section III-A and the projected guidance proposed in Section III-B.

A. Evaluation Protocol, Datasets, and Settings

We follow the VoicePrivacy Challenge 2026 Track 1 (English) evaluation protocol [10] and use the challenge-provided datasets, attack models, and automatic speech recognition (ASR) and speech emotion recognition (SER) evaluation models without modification. Speaker privacy is measured by equal error rate (EER) under lazy-informed and semi-informed attacks, where a higher value indicates stronger protection against speaker re-identification. Linguistic-content preservation is measured by word error rate (WER, lower is better), and emotion preservation is measured by unweighted average recall (UAR, higher is better).

The random-target pool introduced in Section III contains one short utterance (3–6s) from each of 500 speakers in the LibriTTS train-other-500 subset [37]. For every source utterance used by a random-target-conditioned system, one target utterance is sampled from this pool with replacement. The sampled target utterance provides the narrow-band context and wide-band prefix used to construct the AR-generated part

μ_{AR} . The token-prompt part μ_{prm} remains source-derived so that it stays aligned with the source acoustic prompt. A fixed random seed is used, and the same source–target assignment is retained across all projection modes and scale settings to ensure a controlled comparison.⁶

B. Results

Verifying the Roles of the Two Guidance Scales. We take the subset *Libri_dev* to verify whether the two coefficients in Eq. (11) have the intended effects. All configurations use the same source utterances. For random-target-conditioned configurations, the same fixed source–target pairs are used throughout the sweep. Only the AR-generated part μ_{AR} changes with the sampled target; μ_{prm} remains source-derived in all configurations. Privacy is measured on this subset using the dissimilarity score

$$A = 1 - \frac{1}{N} \sum_{i=1}^N \cos(\mathbf{e}_i^{\text{src}}, \mathbf{e}_i^{\text{anon}}), \quad N = 30,$$

where the embeddings are extracted using the same speaker model as the VPC ASV pipeline. A higher value indicates that the anonymized speech is farther from the source speaker. Utility is measured by corpus-level WER over the same 30 utterances using the VPC ASR model.

Figure 2 varies one coefficient while keeping the other fixed.⁷ As expected, increasing λ_s raises the dissimilarity score, although excessive suppression also increases WER. Increasing λ_c , in contrast, lowers WER by strengthening token guidance, but it also reduces the dissimilarity score. These opposite trends confirm the roles assigned to the two terms: λ_s mainly controls source-speaker suppression, whereas λ_c mainly controls linguistic-content preservation.

Guidance Scale Grid Search on *Libri_dev*. After verifying the roles of the two scales, we use the same 30-utterance subset to select the operating point of each system. For each of the three projection modes (Global, Smooth-21, and Token), we test both target-free AR construction and random-target-conditioned AR construction. In both cases, the token-prompt

⁵The screening subset contains 30 speakers, with one long and clearly spoken utterance selected per speaker. It is used only for quick screening; empirically, the selected settings generalize well to the full development and evaluation sets, although tuning on the full development set could potentially yield better settings and would make the dense (λ_s, λ_c) sweep substantially more expensive.

⁶We also performed preliminary checks with several alternative random seeds and source–target assignments and observed only small changes. These checks suggest that using an unrelated target context is more important than the exact target identity for weakening source-speaker information in μ_{AR} .

⁷In Fig. 2, $\lambda_s = 3$ is used as the upper sweep bound for diagnostic analysis, not as the selected operating point.

TABLE I

DEVELOPMENT-SET ABLATION RESULTS. RT MEANS THAT μ_{AR} USES RANDOM-TARGET CONTEXT. HIGHER EER AND UAR ARE BETTER, WHILE LOWER WER IS BETTER. THE HIGHEST EER IS SHADED.

System	RT	λ_s	λ_c	EER	WER	UAR
Built-in		0.7	0.7	33.15	4.18	41.85
Global		0.9	2.8	33.90	4.20	39.21
Smooth-21		1.0	3.0	34.20	4.39	40.35
Token		0.2	2.4	33.48	4.22	40.30
Global	✓	0.9	2.8	34.44	4.72	38.62
Smooth-21	✓	1.0	3.0	34.79	4.47	39.06
Token	✓	0.2	2.4	33.80	4.60	39.43
Global	✓	1.2	2.9	34.61	4.42	40.13
Smooth-21	✓	1.1	2.8	35.09	4.52	40.38
Token	✓	0.1	1.3	34.72	4.36	39.95

TABLE II

TRACK 1 COMPARISON WITH THE OFFICIAL BASELINES. PRIVACY IS MEASURED BY EER (%) UNDER LAZY-INFORMED (L) AND SEMI-INFORMED (S) ATTACKS, AND UTILITY BY WER (%) AND UAR (%). BUILT-IN ANON. USES $(\lambda_s, \lambda_c) = (0.7, 0.7)$ AND SMOOTH-21 USES $(1.1, 2.8)$; † MARKS SYSTEMS USING RANDOM-TARGET CONTEXT FOR μ_{AR} . THE MAXIMUM VALUE IN EACH EER COLUMN IS SHADED.

System	Development				Evaluation			
	L-EER	S-EER	WER	UAR	L-EER	S-EER	WER	UAR
Orig.	7.34	–	1.80	69.08	3.41	–	1.84	71.06
B2	8.15	4.23	10.44	55.66	5.62	3.72	9.96	53.49
B3	26.03	16.32	4.31	38.08	19.06	14.24	4.31	35.24
B4	19.73	15.28	6.16	41.97	14.64	11.13	5.90	42.78
B5	30.94	20.05	4.91	40.08	25.65	17.05	4.44	38.25
Built-in	33.15	24.64	4.18	41.85	28.19	17.78	4.58	40.38
Smooth-21†	35.09	25.14	4.52	40.38	29.39	21.51	4.46	39.57

condition μ_{prm} is extracted from the source utterance. This gives six configurations in total.

For every configuration, both λ_s and λ_c are swept from 0 to 3 with a step size of 0.1. The source utterances and, when applicable, their random-target assignments remain fixed throughout the search. We first retain configurations with WER below 5%, close to the baselines, and then choose the one with stronger source-speaker dissimilarity. The 5% threshold is an application-dependent design choice. The selected values are reported in Table I. After selection, all scale pairs are fixed and no further tuning is performed on the complete development or evaluation sets.

Table I reports the development-set ablation results, while Table II compares the selected Smooth-21 system with the official baselines and built-in anonymization.

Effect of Random-Target Conditioning. Table I isolates the effect of the Stage 1 random-target-conditioned construction of μ_{AR} introduced in Section III-A. To provide a controlled comparison, we first keep (λ_s, λ_c) exactly the same as in each target-free counterpart and change only the use of random-target (RT) conditioning. Under this fixed-scale setting, development Lazy-EER increases from 33.90% to 34.44% for Global, from 34.20% to 34.79% for Smooth-21, and from

33.48% to 33.80% for Token. Thus, all three projection modes achieve higher privacy with comparable utility when RT conditioning is the only changed factor.

When the guidance scales are further tuned for the RT systems, Lazy-EER increases to 34.61%, 35.09%, and 34.72% for Global, Smooth-21, and Token, respectively. These results confirm that the benefit of random-target conditioning is not caused by different hyperparameter choices: introducing an unrelated target context itself consistently reduces source-speaker dependence in the AR-generated part μ_{AR} . We therefore retain random-target-conditioned AR construction in the selected system used for the subsequent semi-informed comparison.

Effect of the Projection Method. Table I also evaluates the projected speech-generation rule introduced in Section III-B. The three configurations without random-target context use the same target-free construction of μ_{AR} and the same source-derived μ_{prm} as the built-in Seed-VC anonymization mode. Their main difference is therefore in Stage 2. The built-in mode uses the complete token condition with prompt and style set to zero, whereas the proposed systems additionally use source prompt and style to estimate the source-conditioning direction and then remove the aligned part of token guidance.

All three projection variants improve development Lazy-EER over the built-in mode: Global, Smooth-21, and Token reach 33.90%, 34.20%, and 33.48%, respectively, compared with 33.16% for built-in anonymization. Their WER remains close to that of the built-in system. This indicates that the privacy gain is not produced only by random-target conditioning, suppressing the source-aligned component during speech generation is itself beneficial. Smooth-21 gives the strongest privacy result among the three projection granularities.

Overall Comparison with Baselines. Table II compares the selected random-target-conditioned Smooth-21 system with the official baselines and the built-in Seed-VC anonymization mode. On the evaluation set, Smooth-21 reaches a Lazy-EER of 29.39% and a Semi-EER of 21.51%, both higher than the corresponding results of the built-in anonymization mode. It also exceeds the strongest official baseline B5 in both evaluation EER measures, while maintaining a comparable WER. Smooth-21 also achieves the highest development EERs, confirming its effectiveness against a semi-informed attacker while preserving competitive utility.

V. CONCLUSION AND FUTURE WORK

We presented a training-free, two-stage voice anonymization method built on Seed-VC v2. The ablation results show that random-target-conditioned AR construction reduces source-speaker dependence in μ_{AR} , while source-conditioning-aware projected CFG further suppresses source-aligned information carried by the complete token guidance during speech generation. Together, the two stages improve speaker privacy while maintaining competitive linguistic and emotion utility on VPC 2026 Track 1. The current study is limited to English anonymization. Future work will extend the method to VPC 2026 Track 2 and investigate its performance in multilingual anonymization.

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